

Cover Letter — TMLR Submission (with arXiv as fallback) To:

TMLR Action Editor From: AGI Research Kit Contributors Date: 2026-08-01 Re: Single-paper submission (replaces a prior 5-paper bundle that has been archived) — Dear Editor, We are pleased to submit our manuscript, "AGI Kit: An End-to-End Self-Improving Tool-Use Pipeline on Consumer Hardware — Empirical Observations", for consideration at TMLR. The submission is a single unified preprint; a prior 5-paper bundle has been archived at [papers/_deprecated/](#) and may be inspected for audit but is no longer the submission of record. The paper reports what we built and what we measured: a four-layer self-improving tool-use pipeline (per-step reflection, semantic strategy memory, continual learning with an A/B safety gate, and bounded recursive self-modification) running entirely on consumer hardware (CPU-only, ~5 GB RAM) with a Qwen3-1.7B base model. Headline numbers (Round 7 ablation, n=8-20 episodes per configuration): Layer ablation: all 5 configs score 77.8% on synthetic GAIA2 mini (eval saturates). On the harder 20-task arithmetic eval, bare scores **5.0%** and full L1-L4 scores **100.0%** - a **+95 percentage point** delta attributable to the layers (Section 4.1). Continual-learning variance: **60.4% +/- 3.6%** across 3 seeds x 15 episodes. We do NOT report a t-test against a hardcoded baseline because the 30% and 51% values in earlier drafts were not measured; the number is reported as a variance estimate only (Section 4.3). A/B safety gate: **12/12** adversarial boundary tests passed (Section 6.1).